

1. Introduction

- a. About me
- b. About you
- c. Class rep
- d. Rules: no rules; basic operating “agreements”
 - i. Respect for each other
 - ii. Contribute, if you know— there may be a “reward”
 - iii. Stop me if you need an explanation, have a question or contribution, etc.
 - iv. It is your responsibility to find out about previous lectures if you were absent. Don’t take us back.

2. Required background (includes the following):

- a. Programming (Object-oriented programming; any programming language)
- b. Data access (databases, json, xml, etc.)
- c. Networks
- d. Distributed applications and processing
- e. Web design
- f. Scripting
- g. Research skills

3. Development platform: Java

- Install current JDK and IDE (e.g., Eclipse, Netbeans, etc.)
- Set up IDE

4. Mode of delivery

- a. Lectures (according to Time table)
- b. Lab and tutorial (**independent** work)
- c. Consultation: 1 hour a week (**Office**: Thursday at 10:00)

Note: no unscheduled consultations apart from c.

5. Communication

- a. Moodle
- b. Formal email

6. Textbook and resources: no prescribed textbook

- a. Java books (or other resources) which cover
 - Programming fundamentals
 - Inheritance, polymorphism, interfaces
 - Java and XML
 - Java security
 - Java Server Pages
- b. Handouts

7. Plagiarism (refer to Study guide)

8. Groups and group work

9. Topics (as specified in Module Descriptor)

- 1. Intersystem Communications
 - a. Architectures for integrating systems
 - b. DCOM, CORBA, RMI
 - c. Web services and middleware
 - d. Network programming
 - e. Message and queuing services
 - f. Low-level data communications
- 2. Data Mapping and Exchange
 - a. Metadata
 - b. Data representation and encoding
 - c. XML, DTD, XML schemas
 - d. Parsing XML documents
 - e. XSL, XSLT, and XPath
- 3. Integrative Coding
 - a. Design patterns
 - b. Interfaces
 - c. Inheritance
- 4. Scripting Techniques

- a. Scripting and the role of scripting languages
- b. Creating and executing scripts
- c. Influence of scripting on programming
- 5. Software Security Practices
 - a. Evidence-based security vs. code access security
 - b. Best security coding practices
 - c. Authentication to system resources and services
 - d. Encryption of data between systems and services

Note: Topic 5 is self-study equivalent to about one week (5 hours)

10. Outcomes

1. Describe and contrast the different types of architectures for integrating systems.
2. Design, develop and test a program that uses a messaging service
3. Design, develop and test a program that uses XML
4. Design, develop and test a program that applies design patterns to solve a problem.
5. Write, debug and test an interactive web-based application
6. Perform a security audit of the code of an existing system

11. Activity and time allocation

Learning activity	% learning time
Contact (Lectures)	40
Practical	20
Other assignments	20
Independent study	20